

Offline Lead Capture

Collect student leads at an event with no internet, then upload them all with one button when you're back online.

Live at **Students & Pipeline** → **Offline Capture** (</students/offline>).

Screenshots below are of the real system, captured against production.

For ICRs: using it at an event

Setting a PIN

The first time you open Offline Capture, you'll be asked to set a PIN of at least six digits. Leads held on the device are encrypted with it, so a lost phone doesn't mean lost student data.

Nobody can recover this PIN — not IT, not us. If you forget it, leads still waiting on the device can't be read and have to be erased. Upload at the end of each day and there's nothing to lose.

You'll be asked for it each time you open the page.

Before you travel — do this on wifi

Open **Students & Pipeline**, then **Offline Capture**.

- Dashboard
- Clients
- Students & Pipeline**
- Sources
- Markets
- Stakeholders
- Activities
- Events
- Analytics
- Reports
- Tasks
- HR & ERP
- Travel
- Risk & Compliance

SD Screenshot Demo

shot-icr@illumestudentservices...

ICR

Sign Out

Illume Student Advisory Services

Search students, institutions...

SD Screenshot Demo

Home > Students

Student Pipeline

Track and manage student leads through the recruitment funnel.

Offline capture holds up to 100 leads per session. Upload them before collecting more — beyond 100 nothing further can be saved on the device.

Kanban

List

Export

Offline Capture

Add Lead

Search leads...

All Stages

All Institutions

2 leads

NEW LEAD 2

Tunde Bello

NG Nigeria

MSc Data Science PG

Jan 2027 0d

SD

Amara Okeke

NG Nigeria

BSc Computer Scie... UG

Sep 2027 0d

SD

CONTACTED 0

Drop cards here

QUALIFIED 0

Drop cards here

APPLICATION SUBMITTED

Drop cards here

The first time you open it, it will tell you it isn't ready:

- Dashboard
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- Students & Pipeline**
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- Markets
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- Activities
- Events
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< Collapse

SD Screenshot Demo

shot-icr@illumestudentservices...

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Illume Student Advisory Services

SD Screenshot Demo

Home > Students > Offline Capture

Offline Capture

Collect leads at an event with no internet, then upload them all at once.

Connected

Capturing as Screenshot Demo

Prepare for offline

Upload all

Not ready for offline yet. Tap "Prepare for offline" while you still have a connection, or the Source, Event and Institution lists will be empty at the booth — and Lead source is required before a lead can progress.

Offline capture holds up to 100 leads per session. Upload them before collecting more — beyond 100 nothing further can be saved on the device.

New lead

Both an email and a phone number are needed — ask for both while the student is still with you.

First name *

Last name *

Email *

Phone *

Citizenship *

Country of residence *

Intended programme *

Study level *

Select

Intake month *

Select

Intake year *

Intended destination

Preferred country

Tap **Prepare for offline**. This downloads the Lead Source, Event and Institution lists onto your device, and installs the page so it opens without a signal.

Page 3 of 13

Illume Student Advisory Services

Search students, institutions...

SD Screenshot Demo

Home > Students > Offline Capture

Offline Capture

Collect leads at an event with no internet, then upload them all at once.

Connected Capturing as Screenshot Demo

[Prepare for offline](#) [Upload all](#)

Lists last downloaded 02/08/2026, 19:52:52 · 15 sources, 0 events, 8 institutions

Offline capture holds up to 100 leads per session. Upload them before collecting more — beyond 100 nothing further can be saved on the device.

New lead

Both an email and a phone number are needed — ask for both while the student is still with you.

First name *

Last name *

Email *

Phone *

Citizenship *

Country of residence *

Intended programme *

Study level *

Intake month *

Intake year *

Intended destination

Preferred country

Ready for offline
Lists saved to this device.

You'll see when the lists were last downloaded. **Do this the day you travel** — if the lists are weeks old, an event created since then won't be in the list.

This step is not optional. Without it, Lead Source is empty at the booth — and Lead Source is one of the fields required before a lead can move past "New Lead". You'd capture students who then can't progress.

At the booth

Fill in the form and tap **Save to device**. You need **both an email and a phone number** — ask for both while the student is still with you.

If the event issues scannable badges, tap **Scan badge** and hold it in the frame. Whatever it can read is shown for you to confirm before it fills the form, and you can still change anything. Badges differ between organisers, so check what came through — some carry a full contact card, others only an attendee number.

Scan badge only appears on Android and Chrome. iPhones can't do this yet; type the details in as normal.

The screenshot shows the Illume Student Advisory Services interface. On the left is a dark blue sidebar with navigation links: Dashboard, Clients, Students & Pipeline (highlighted), Sources, Markets, Stakeholders, Activities, Events, Analytics, Reports, Tasks, HR & ERP, Travel, and Risk & Compliance. At the bottom of the sidebar is a user profile for 'Screenshot Demo' with email 'shot-icr@illumestudentservices...' and a 'Sign Out' button. The main content area has a header with the title 'Illume Student Advisory Services', a search bar 'Search students, institutions...', and a notification bell. Below the header is a form with several fields: 'Intended destination' (Canada), 'Preferred country' (empty), 'Lead source' (None), 'Event' (None), 'Institution' (None), 'Budget range' (Not asked), and 'English status' (Not asked). There is a 'Notes' section with a text area containing 'Anything worth remembering about this conversation'. Below the notes is a section titled 'May we email them?' with a red asterisk and a note: 'Canadian anti-spam law requires permission before sending marketing email. Ask the student directly — leaving this unanswered means we cannot email them.' There are three buttons: 'Yes, they agreed', 'No, they declined', and 'Didn't ask'. At the bottom right of the form is a '+ Save to device' button. Below the form is a section titled 'Held on this device (0)' with a 'Refresh' button and a note: 'Nothing captured yet. Leads saved here stay on the device until you upload them.'

Note the question at the bottom: **"May we email them?"** Canadian anti-spam law requires permission before sending marketing email. There are three answers, and they mean different things:

Answer	Meaning
Yes, they agreed	You may email them
No, they declined	You must not email them
Didn't ask	Nobody asked — you can still ask later

"Didn't ask" is not the same as "declined". Leave it honest.

Each lead is saved onto the phone:

- Dashboard
- Clients
- Students & Pipeline**
- Sources
- Markets
- Stakeholders
- Activities
- Events
- Analytics
- Reports
- Tasks
- HR & ERP
- Travel
- Risk & Compliance

SD

Screenshot Demo

shot-icr@illumestudentservices...

ICR

Sign Out

Illume Student Advisory Services

Search students, institutions...

SD Screenshot Demo

Home > Students > Offline Capture

Offline Capture

Collect leads at an event with no internet, then upload them all at once.

Connected

Capturing as Screenshot Demo

Prepare for offline

Upload (1)

Lists last downloaded 02/08/2026, 19:52:52 · 15 sources, 0 events, 8 institutions

Offline capture holds up to 100 leads per session. Upload them before collecting more — beyond 100 nothing further can be saved on the device.

New lead

Both an email and a phone number are needed — ask for both while the student is still with you.

First name *	Last name *
Nkechi	Obi
Email *	Phone *
Citizenship *	Country of residence *
Nigerian	Nigeria
Intended programme *	Study level *
BSc Computer Science	Select
Intake month *	Intake year *
Select	2027
Intended destination	Preferred country
Canada	

Saved to this device

1 of 100 held. Not yet uploaded.

With no signal

Everything works the same. The badge at the top changes to **No connection**, and **Upload** is disabled until you're back online.

Dashboard

Clients

Students & Pipeline

Sources

Markets

Stakeholders

Activities

Events

Analytics

Reports

Tasks

HR & ERP

Travel

Risk & Compliance

< Collapse

SD Screenshot Demo

shot-icr@illumestudentservices...

ICR

Sign Out

Illume Student Advisory Services

Search students, institutions...

SD Screenshot Demo

Home > Students > Offline Capture

Offline Capture

Collect leads at an event with no internet, then upload them all at once.

No connection

Capturing as Screenshot Demo

Prepare for offline

Upload (1)

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First name *

Nkechi

Last name *

Obi

Email *

Phone *

Citizenship *

Nigerian

Country of residence *

Nigeria

Intended programme *

BSc Computer Science

Study level *

Select

Intake month *

Select

Intake year *

2027

Intended destination

Canada

Preferred country

Keep capturing. The counter next to Upload shows how many are waiting.

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Dashboard

Clients

Students & Pipeline

Sources

Markets

Stakeholders

Activities

Events

Analytics

Reports

Tasks

HR & ERP

Travel

Risk & Compliance

< Collapse

SD Screenshot Demo

shot-icr@illumestudentservices...

ICR

Sign Out

Illume Student Advisory Services

Search students, institutions...

SD Screenshot Demo

Intended destination

Canada

Preferred country

Lead source

None

Event

None

Institution

None

Budget range

Not asked

English status

Not asked

Notes

Anything worth remembering about this conversation

May we email them?

Canadian anti-spam law requires permission before sending marketing email. Ask the student directly — leaving this unanswered means we cannot email them.

Yes, they agreed

No, they declined

Didn't ask

+ Save to device

Held on this device (2)

Refresh

Amara Okeke

amara.okeke@example.com · +234-80-11122233

02/08/2026, 19:53:27

Tunde Bello

tunde.bello@example.com · +234-80-99988877

Waiting

Saved to this device

2 of 100 held. Not yet uploaded.

Back on wifi

The badge returns to **Connected** and Upload becomes available.

Illume Student Advisory Services

SD Screenshot Demo

Dashboard

Clients

Students & Pipeline

Sources

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Stakeholders

Activities

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Tasks

HR & ERP

Travel

Risk & Compliance

< Collapse

SD Screenshot Demo

shot-icr@illumestudentservices...

ICR

Sign Out

Home > Students > Offline Capture

Offline Capture

Collect leads at an event with no internet, then upload them all at once.

Connected

Capturing as Screenshot Demo

Prepare for offline

Upload (2)

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Nkechi

Last name *

Obi

Email *

Phone *

Citizenship *

Nigerian

Country of residence *

Nigeria

Intended programme *

BSc Computer Science

Study level *

Select

Intake month *

Select

Intake year *

2027

Intended destination

Canada

Preferred country

Tap **Upload**. The device clears and the leads are in the CRM.

Illume Student Advisory Services Search students, institutions...

Home > Students > Offline Capture

Offline Capture

Collect leads at an event with no internet, then upload them all at once.

Connected Capturing as Screenshot Demo Prepare for offline Upload all

Lists last downloaded 02/08/2026, 19:52:52 · 15 sources, 0 events, 8 institutions

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Nigerian	Nigeria
Intended programme *	Study level *
BSc Computer Science	Select
Intake month *	Intake year *
Select	2027
Intended destination	Preferred country
Canada	

Things to know

Up to 100 leads per session. At 100 the form stops accepting new ones until you upload. Not a technical limit — a limit on how much unsent work sits in one place, because a lost phone loses all of it.

Upload before the end of each day if you can. Leads only exist on that one device until they're uploaded.

If the upload is interrupted, just press it again. Every lead carries a hidden ticket number, so leads that already arrived are recognised and skipped rather than created twice.

One bad record won't block the rest. Each lead is sent separately. Failures stay on the device with a note explaining what's wrong. Tap the pencil next to a rejected lead to load it back into the form, fix it, and save — it goes up with the next upload. Everything else uploads normally.

Nothing is deleted from the device until the server confirms it. If the upload fails, nothing is removed and you'll be told so.

If you're signed out mid-upload, sign in and press Upload again. Nothing is lost — sessions expire after 48 hours and an event can outlast that.

Don't use a private/incognito window. Offline storage is blocked there. The page refuses to run rather than appearing to work and losing everything on close.

For developers

Where things live

Path	Purpose
<code>app/(dashboard)/students/offline/page.tsx</code>	Route
<code>.../offline/_components/offline-capture-client.tsx</code>	The capture screen
<code>.../offline/_components/register-offline-worker.tsx</code>	Installs the service worker
<code>.../offline/_components/pin-gate.tsx</code>	PIN lock in front of the screen
<code>.../offline/_components/badge-scanner.tsx</code>	Camera scanning
<code>lib/offline-queue.ts</code>	IndexedDB queue, encrypted
<code>lib/offline-crypto.ts</code>	AES-GCM + PBKDF2
<code>lib/badge-scan.ts</code>	Badge payload parsing
<code>lib/offline-capture.ts</code>	The 100 limit and its warning text
<code>lib/lead-options.ts</code>	Dropdown options, shared with the online form
<code>app/api/leads/offline-sync/route.ts</code>	Batch upload
<code>app/api/leads/offline-reference/route.ts</code>	Dropdown data download
<code>public/sw.js</code>	Service worker
<code>prisma/manual/010-offline-capture.sql</code>	<code>Lead.captureId</code>
<code>prisma/manual/011-marketing-consent.sql</code>	Consent columns

Design decisions worth not undoing

`Lead.captureId` is a **unique idempotency key**, generated on the device at capture. The upload endpoint checks it before inserting, and the unique index backs that up against a race. The existing duplicate detection only *flags* `isDuplicate` after the fact — it does not prevent the row. Without `captureId`, a retried upload creates every already-delivered lead a second time.

The upload endpoint is deliberately not one transaction. A batch is a hundred independent students; wrapping them together means one malformed row discards ninety-nine good ones. Each is attempted alone and reported individually as `created` / `already_synced` / `failed`.

The device deletes only what the server confirms. Anything not mentioned in the response stays put. The device is the only copy.

Validation on the device mirrors the server exactly, both email and phone included. A looser check would queue leads that can only fail on upload hours later, with the student long gone.

marketingConsent is Boolean?, three-valued. `NULL` means nobody asked; `false` means asked and declined. A boolean defaulting to `false` collapses those, and under CASL that distinction is the entire point. Migration 011 does not back-fill for the same reason.

The service worker is deliberately narrow. It touches the capture page document and `/_next/static/*` and nothing else — never `/api/`, never other navigations, never non-GET. It caches only 200 responses, so it cannot pin the `/login` redirect in place of the page. It is registered from the capture page alone, not the root layout, so it installs only for people who attend events.

Kill switch. A service worker lives on the device and keeps misbehaving after you revert the deploy that caused it. To disable, replace the body of `public/sw.js` with:

```
self.addEventListener("install", () => self.registration.unregister());
```

and deploy. Devices pick it up on their next visit.

proxy.ts must keep excluding sw.js from its matcher. The service worker spec rejects a registration whose script request redirects, and every other path through the proxy redirects a signed-out visitor to `/login`. Left in, the worker silently fails to install with nothing in any server log — the request returns an ordinary 307. Re-check with:

```
curl -sI https://illumestudentservices.cloud/sw.js | head -1 # must be 200
```

Don't join template literals with + in lib/offline-capture.ts. The limit is a module-local constant, so the minifier folds `${OFFLINE_CAPTURE_LIMIT}` to `100` — and while folding two `+`-joined literals it drops the static text between the interpolation and the closing backtick. The banner shipped reading *"holds up to 100Upload them before collecting more"*. TypeScript, eslint and the build all passed. Keep it as one literal.

Encryption at rest

Held leads are encrypted with AES-GCM under a key derived from the device PIN (PBKDF2-HMAC-SHA256, 310,000 iterations — OWASP's floor, and roughly 100ms on a mid-range phone). `captureId`, timestamps and status stay in the clear so the queue can be counted, sorted and reconciled without a key; none of them says anything about a student.

The threat is narrow and worth stating: a phone left on a stand or taken from a bag. It does not defend against malware running as the user, and cannot — the key is derived in the browser that would be compromised. A six-digit PIN is weak, which is why the derivation is deliberately expensive: brute-forcing the space costs hours of work on the device rather than seconds, and the queue is meant to be emptied daily.

There is no PIN recovery. The key exists only in the PIN. A forgotten PIN means the held leads cannot be read, and the only way forward is to erase them. That is offered explicitly, with a warning, because the alternative is a device that can never capture again. Anything that could recover the key would defeat the one threat this exists for.

Records written before encryption existed are discarded rather than carried over — they cannot be encrypted retroactively with a key that did not exist when they were written, and leaving readable copies beside encrypted ones would make the encryption decorative. The gate warns before that happens.

Badge scanning

`lib/badge-scan.ts` reads what organisers actually put on badges: vCard, MECARD, JSON, a URL with query parameters, `Key: value` lines, or unstructured text. There is no standard, so it fills only the fields it is confident about and leaves the rest to the ICR. A wrong email captured at a booth is worse than a blank one, because nobody goes back to check — so **no name is ever guessed from loose text**, only from a recognised structure.

Scanning uses the browser's native `BarcodeDetector`. Chrome and Android have it; Safari and iOS do not. The button is hidden where it is unsupported rather than failing when pressed. Everything read is shown for confirmation before it touches the form.

Correcting a rejected lead

A failed capture keeps its reason and can be loaded back into the form, fixed and resaved. **The `captureId` is preserved through a correction** — a corrected lead is the same lead, and issuing a fresh one would let the original attempt and the retry both land. The 100-lead ceiling does not block a correction, since it replaces a record already counted.

Known gaps

- **iPhone cannot scan badges.** Safari has no `BarcodeDetector`. A JS decoder library would close this at the cost of a dependency and a slower scan.
- **The PIN is not rate-limited.** Attempts are unlimited; the cost of the key derivation is the only brake.
- **Reference lists do not expire.** The screen shows when they were downloaded but will not stop you using month-old data.